

L13: Lesson Summary

This lesson focused on the use of statistical methods in the analysis of network data. This is a broad area, with many different topics. We mostly focused on two of them:

a) network sampling: design of different sampling strategies and inference of network properties from those samples,

b) topology inference: detecting the presence of missing links, creating association networks, and using tomography methods to discover the topology of the network.

Here is a list of other topics in the statistical analysis of network data:

- Statistical modeling and prediction of dynamic processes on networks (i.e., applying statistical methods such as Markov Random Fields or Kernel-Based Regression on dynamic processes on networks such as epidemics)
- Analysis and design of directed and undirected graphical models
- Efficient algorithms for the computation of network motifs from sampled network data
- Efficient algorithms for the computation of centrality metrics and communities.

In parallel, some of these problems are also pursued by the Machine Learning community, as we will see in the next lesson.